

1. BEST APPROXIMATING AFFINE SPACE

As discussed in class, we now see the proof of the following Theorem. Some notations are in order before the statement is presented.

S : A subspace of the vector space $\mathbb{R}^p$ with $d(S) = k (\leq p)$.

$\underline{a}$: A vector in $\mathbb{R}^p$.

$\underline{a} + S$: An affine space.

$\hat{x} = \underline{a} + P_S(x - \underline{a})$: Projection of x onto the affine subspace $\underline{a} + S$. Here P_S is the orthogonal projection matrix onto S . Note that if the columns of $A_k = [\underline{w}_1 \cdots \underline{w}_k]$ forms an orthonormal basis of S , then $P_S = A_k A_k^T = \sum_{i=1}^k \underline{w}_i \underline{w}_i^T$ where $A_k^T A_k = I_k$.

Let S_k^* be the subspace generated by the first k eigen-vectors $\underline{v}_1, \dots, \underline{v}_k$. As these vectors are orthonormal, the orthogonal projection matrix onto S_k^* is $P_k^* = \sum_{i=1}^k \underline{v}_i \underline{v}_i^T$.

Let $\underline{X}_{p \times 1}$ be a random vector. Then, $d(\underline{a} + S, \underline{X}) = \mathbb{E}(\|\underline{X} - \hat{\underline{X}}\|^2)$.

Theorem 1. Among all k -dimensional subspaces S of $\mathbb{R}^p$ and all vectors in $\underline{a} \in \mathbb{R}^p$, $d(\underline{a} + S, \underline{X})$ is minimized when $\underline{a} = \mu_X$ and $S = S_k^*$.

Proof. We start by proving the following lemma.

Lemma 1. Let Σ be covariance matrix of $\underline{X}$. Then

$$\max_{\{P | P \text{ is an orthogonal projection matrix with } \rho(P)=k\}} \text{tr}(\Sigma P) = \sum_{i=1}^k \lambda_i$$

where λ_i s are first k eigenvalues of Σ , and the maximum is achieved by P_k^* .

Proof. As P is an orthogonal projection matrix with $\rho(P) = k$, we may write $P = A_k A_k^T$ where $A_k^T A_k = I_k$. Thus,

$$\begin{aligned} \text{tr}(\Sigma P) &= \text{tr}(V \Lambda V^T A_k A_k^T) \\ &= \text{tr}\left(\sum_{i=1}^p \lambda_i \underline{v}_i \underline{v}_i^T A_k A_k^T\right) \\ &= \left(\sum_{i=1}^p \lambda_i \underline{v}_i^T A_k A_k^T \underline{v}_i\right) \\ &= \sum_{i=1}^p \lambda_i \beta_i \end{aligned}$$

where $\beta_i = \underline{v}_i^T A_k A_k^T \underline{v}_i = (A_k^T \underline{v}_i)^T (A_k^T \underline{v}_i) \geq 0$ for all $i = 1(1)p$. Further, for $i \in \{1, \dots, p\}$,

$$\begin{aligned}
\beta_i &= \underline{v}_i^T A_k A_k^T \underline{v}_i \\
&= \frac{\underline{v}_i^T A_k A_k^T \underline{v}_i}{\underline{v}_i^T \underline{v}_i} && (\text{as } \underline{v}_i^T \underline{v}_i = 1) \\
&\leq \lambda_{\max}(A_k A_k^T) && (\text{where } \lambda_{\max}(A_k A_k^T) \text{ is the maximum eigenvalue of } A_k A_k^T) \\
&= \lambda_{\max}(A_k^T A_k) && (\text{as the non-zero eigenvalues of } AB \text{ and } BA \text{ are the same}) \\
&= 1
\end{aligned}$$

and

$$\begin{aligned}
\sum_{i=1}^k \beta_i &= \sum_{i=1}^p \underline{v}_i^T A_k A_k^T \underline{v}_i \\
&= \text{tr}(V^T A_k A_k^T V) \\
&= \text{tr}(A_k A_k^T V V^T) \\
&= \text{tr}(A_k A_k^T) \\
&= \text{tr}(A_k^T A_k) \\
&= k.
\end{aligned}$$

Thus, $\text{tr}(\Sigma P) \leq \sum_{i=1}^p \lambda_i \beta_i$ where for each $i = 1(1)k$, $\beta_i \in [0, 1]$ and $\sum_{i=1}^p \beta_i = k$. Now, given $\lambda_1 \geq \dots \geq \lambda_p$, $\text{tr}(\Sigma P)$ is maximized when $\beta_j = 1$ for all $j = 1(1)k$ and $\beta_j = 0$ for all $j = k+1(1)p$. Hence, $\text{tr}(\Sigma P) \leq \lambda_1 + \dots + \lambda_k$.

If $P = P_k^*$ then for $j = 1(1)p$, $\beta_j = \underline{v}_j^T \{\sum_{l=1}^k \underline{v}_l \underline{v}_l^T\} \underline{v}_j = \sum_{l=1}^k \underline{v}_j^T \underline{v}_l \underline{v}_l^T \underline{v}_j$. This gives us

$$\beta_j = \begin{cases} 1 & \text{if } j = 1(1)k, \\ 0 & \text{if } j = k+1(1)p. \end{cases}$$

Therefore, $\text{tr}(\Sigma P)$ is maximized when $P = P_k^*$. ■

Corollary 1. $\text{tr}(\Sigma(I - P)) \geq \text{tr}(\Sigma(I - P_k^*))$ where P and P_k^* are as in the previous result.

We now complete the proof of the theorem.

$$\begin{aligned}
d(\underline{a} + S, \underline{X}) &= \mathbb{E}(\|\underline{X} - \hat{\underline{X}}\|^2) \\
&= \mathbb{E}(\|(I - P_S)(\underline{X} - \underline{a})\|^2) \\
&= \mathbb{E}(\|Q_S[(\underline{X} - \underline{\mu}_X) + (\underline{\mu}_X - \underline{a})]\|^2) \quad (\text{where } I - P_S = Q_S, \text{ is another projection matrix}) \\
&= \mathbb{E}(\|Q_S(\underline{X} - \underline{\mu}_X)\|^2) + \mathbb{E}(\|Q_S(\underline{\mu}_X - \underline{a})\|^2) + 2\mathbb{E}((\underline{\mu}_X - \underline{a})Q_S^T Q_S(\underline{X} - \underline{\mu}_X)) \\
&= \mathbb{E}(\|Q_S(\underline{X} - \underline{\mu}_X)\|^2) + \mathbb{E}(\|Q_S(\underline{\mu}_X - \underline{a})\|^2) \\
&\geq \mathbb{E}(\|Q_S(\underline{X} - \underline{\mu}_X)\|^2) \\
&= \mathbb{E}((\underline{X} - \underline{\mu}_X)^T Q_S^T Q_S (\underline{X} - \underline{\mu}_X)) \\
&= \mathbb{E}((\underline{X} - \underline{\mu}_X)^T Q_S (\underline{X} - \underline{\mu}_X)) \\
&= \mathbb{E}(\text{tr}(Q_S \underline{X} - \underline{\mu}_X)^T (\underline{X} - \underline{\mu}_X)) \\
&= \text{tr}(Q_S \mathbb{E}((\underline{X} - \underline{\mu}_X)^T (\underline{X} - \underline{\mu}_X))) \\
&= \text{tr}(Q_S \Sigma) \\
&= \text{tr}((I - P_S) \Sigma) \\
&\geq \text{tr}((I - P_k^*) \Sigma).
\end{aligned}$$

In Line (4) above,

$$\begin{aligned}
\mathbb{E}((\underline{\mu}_X - \underline{a})Q_S^T Q_S(\underline{X} - \underline{\mu}_X)) &= \mathbb{E}((\underline{\mu}_X - \underline{a})Q_S(\underline{X} - \underline{\mu}_X)) \\
&= ((\underline{\mu}_X - \underline{a})Q_S \mathbb{E}(\underline{X} - \underline{\mu}_X)) \quad (\text{as } Q_S^T Q_S = Q_S, \text{ being a projection matrix}) \\
&= 0
\end{aligned}$$

Therefore, the result follows. ■

REMARK 1.1. The projection of $\underline{X}$ by inserting the optimal choices $\underline{a} = \underline{\mu}_X$ and $S = S_k^*$, we have $\hat{\underline{X}}^* = \underline{\mu}_X + P_k^*(\underline{X} - \underline{\mu}_X) = \underline{\mu}_X + \sum_{i=k}^k \underline{Y}_i \underline{v}_i$ where $\underline{Y}_i$ is the i -th PC.